Echoes of Ideology: Evaluating Political Bias and Stability in Large Language Models Through Quantitative Analysis and Standardized Typology Testing

Abstract

This study measures political bias and response consistency across 19 large language models from 11 organizations, using two standardized political-typology instruments: the Political Compass and 8Values. Each model was queried 60 times per instrument (2,280 total administrations) via the OpenRouter API, with both instruments scored against authoritative sources rather than an approximate cross-test formula. Every model leaned the same direction on every axis: economically left and socially libertarian on the Political Compass, and toward equality, peace, liberty, and progress on 8Values. Classical ANOVA, Welch's ANOVA, and Kruskal-Wallis tests agreed at $p < .0001$ on every axis, with large effect sizes ($\eta^2 = 0.56$ to 0.73), though the magnitude of lean varied substantially by model. Trial-to-trial consistency, measured with a range-normalized Ideological Stability Score, varied by specific model rather than by provider, country, or model size. All code, raw trial data, and analysis scripts are released at https://github.com/sletch1/AI-Political-Bias-Research-Paper.

1 Introduction

This study considers the cognitive implications of AI beyond automation. Technological advancements are often viewed as tools that automate labor, improving daily life efficiency. Such an assumption is based on historical precedent. The rise of inventions such as the steam engine, which mechanized power for factories, trains, and ships, revolutionized transportation and production. Similarly, the invention of the spinning Jenny transformed textile production. This trend continues into modern day. Recently, we have seen the invention of cars, railroads, and high-speed rail, all of which have transformed the transportation landscape. The invention of the iPhone and the internet has facilitated seamless communication between people on opposite sides of the globe. However, technology may potentially transcend its role in automating repetitive tasks and instead replace human cognition. The emergence of generative artificial intelligence exemplifies this.

Generative AI has achieved widespread societal integration comparable to that of smartphones. Similar to how almost 9 out of 10 teenagers in the US own an iPhone, more than 4 out of 5 students have used generative artificial intelligence (Pandey, 2021; Kelly, 2024). Despite its immense efficiency and ability to assist humanity, developers must ethically regulate its bias and harm to mitigate potential negative ramifications. With the rise of AI comes moral implications that we must address.

Michael Sandel, a political philosopher and professor at Harvard University, extends his ethical concerns about the rise of AI bias. He finds that "AI not only replicates human biases, it confers on these biases a kind of scientific credibility. It makes it seem that these predictions and judgments have an objective status" (Pazzanese, 2020). A recent concern with the rise of AI bias is bias baked into the knowledge resources AI systems draw on: a 2021 EMNLP study found demographic and representational bias in 38.6% of entries in GenericsKB, a commonsense-knowledge resource used to ground AI systems' factual claims (Mehrabi et al., 2021). Bias in AI systems is not limited to factual grounding: it has also been documented along racial and gender lines, for instance in covertly racist judgments tied to dialect (Hofmann et al., 2024) and in direct measurements of gender and racial bias in LLM outputs (An et al., 2025). Due to the immense focus on misinformation and these other bias categories, political bias in AI tends to be overshadowed.

Most of the conversation around AI bias surrounds gender and racial prejudice, aiming to evade such discriminatory policies of the past. However, political bias is just as detrimental to society. Zvi Mowshowitz from *The New York Times* explains how social media has paved the way for political polarization, as algorithms feed people more content that resonates with them, pushing them further down a particular ideological path (Mowshowitz, 2024). Partisan LLMs exacerbate this trend. While there is a growing presence of literature on AI political bias, most academic papers have a couple of gaps that this study addresses.

The first gap is breadth of model coverage. Most academic studies of political bias in LLMs examine a small, convenience-sampled handful of flagship models, typically from two or three organizations, and are re-run infrequently enough that they miss newer entrants entirely. This study instead evaluates 19 models spanning 11 organizations across four countries, chosen deliberately to cover both closed and open-weight models, multiple size and price tiers within the same model family, and providers outside the usual US-centric set, including DeepSeek and Qwen (China) and Mistral (France).

The second gap is that while many studies examine political bias, few compare political consistency and stability across a broad model set. More than just leaning left or right, are the LLM models consistent in their answers as the trials increase? Do the LLMs give varying answers? Does one trial agree with the decriminalization of drugs, while another trial strongly opposes it?

The third gap is the test comparison. Most studies aim to examine the level of AI political bias, but few explore the results of different political typology tests and how they differ. This analysis will show that the same AI model can result in varying levels of political extremism on different tests.

The research question for this study is: "To what extent do large language models from a broad set of developers exhibit ideological bias and response consistency across political typology assessments, and does either property track a model's developer, country of origin, or size?" This study aims to identify political bias and consistency within models and compare them. It is hypothesized that most models will lean in a similar ideological direction, consistent with prior literature, but that the magnitude of that lean, and the trial-to-trial consistency of a model's answers, will vary by specific model rather than clustering neatly by developer, country, or size. It is further hypothesized that the 8Values test, which offers a 'Neutral/Unsure' option, a larger number of questions, and multiple nuanced factors per axis, will yield less ideological extremity than the Political Compass test.

This research study aims to map ideological leanings, quantify response consistency within a data-collection sitting, and compare the two typology tests to see whether they yield different results for the same model.

1.1 Related Work

The study's foundation is based on Associate Professor David Rozado's 2024 study "The Political Preferences of LLMs" (Rozado, 2024). This study aligns with the interests of scholars in AI, language models, and political bias, sharing a goal similar to Professor Rozado's research but differing in scale, model selection, and methodology. Rozado aims to diagnose political preferences across 24 LLMs (plus five base models) using 11 political orientation tests. In contrast, the current study uses two typology tests across 19 models to measure political bias and within-session response consistency. This study emphasizes the consistency of each model's political bias in addition to its direction. Both studies use models from OpenAI (OpenAI, 2023) and Anthropic (Anthropic, 2023), as well as the Political Compass and 8Values, as tests. A comparably scaled recent study, Buyl et al.'s "Large Language Models Reflect the Ideology of their Creators," probes 19 LLMs (matching this study's model count) using open-ended generation about thousands of political figures rather than a fixed questionnaire (Buyl et al., 2026); the present study's use of two independent, structured typology instruments, each scored against its own authoritative source, is a methodologically different but complementary approach to the same underlying question.

Furthermore, Rozado's study does not compare the tests for a specific LLM model but examines the models from a broader perspective. Both studies score every response against the tests' own authoritative outputs rather than a hand-derived measure, though the automation differs: Rozado utilized automated stance detection (GPT-3.5), validated by human cross-check, whereas the current study drives the Political Compass's own live scoring page directly and ports 8Values' own published scoring algorithm line for line, so that no separate stance-classification step is needed for either instrument. Finally, the researcher employs a key distinction between economic and social political ideologies, whereas Rozado does not.

Another relevant study, conducted by Rettenberger et al., examines political bias from the EU and German perspectives (Rettenberger et al., 2025). Both the author's analysis and Rettenberger et al. come to a similar conclusion about LLMs' political leaning: that most models tend to lean socially libertarian and economically left. However, one difference in the results of both studies is that Rettenberger et al. found that "smaller" models tended to be more neutral. The current study's 19-model roster, which spans multiple size tiers within several model families, did not find a consistent relationship between model size and neutrality. The key novelties between the two studies diverge. In the researcher's analysis, they introduce the Ideological Stability Score (ISS) to quantify response variability; Rettenberger et al. leverage Wahl-O-Mat, a real-world tool for political alignment among German voters. Moreover, Rettenberger et al.'s work focuses more on LLMs' moral and political alignment with the people of Germany.

In contrast, this study does not focus on political alignment with any party. Moreover, Rettenberger et al. compare German and English prompts, while this study is focused on English-based prompts. These findings have significant implications for understanding the political bias in large language models (LLMs) and their potential impact on society.

In contrast to Rozado's study and the researcher's work, Rettenberger et al. incorporate language variation in prompts. However, Rozado and Rettenberger et al. do not emphasize response consistency to the same degree. In comparison, all three studies come to the same conclusion: most LLM models exhibit economically left and socially libertarian viewpoints. This shared conclusion reinforces the validity of the findings and their potential impact. Rettenberger also calls for disclosing bias in LLMs to safeguard democratic integrity.

Another relevant study by PhD candidate Suyash Fulay and Research Scientist Jad Kabbara examines how political bias shifts following the 'truth testing' process on an LLM model (Fulay et al., 2024). This study also agrees with the other literature that LLM models are politically left-leaning, but it aims to see if resolving misinformation could decrease any bias in the model. Fulay et al.'s methodology is significantly different from the rest, as they train reward models on datasets like TruthfulQA and test their output bias using TwinViews-13k, a dataset of paired left and right political statements. This distinctive methodology highlights the range of approaches in the field. One similarity between this analysis and that study is the use of inferential statistical analysis: the current study relies on Welch's ANOVA, the Kruskal-Wallis test, and the Games-Howell post-hoc test as its primary evidence (with the classical one-way ANOVA and Tukey HSD reported alongside for comparison), justified below by explicit assumption checks. In contrast, Fulay et al. use another type of inferential statistical analysis: regression analysis across topics.

Bang et al.'s "Measuring Political Bias in Large Language Models: What Is Said and How It Is Said" (Bang et al., 2024) offers a complementary methodological lens, examining not only which policy positions a model favors but the rhetorical framing used to express them. This study's questionnaire-based approach captures the former, a model's stated position on a fixed set of statements, but, consistent with Bang et al.'s distinction, cannot speak to the latter, since a forced-choice label carries no framing information; that boundary is stated explicitly rather than left implicit.

Two recent papers bear directly on this study's consistency framing and must be reconciled rather than left unacknowledged. Ceron et al.'s "Beyond Prompt Brittleness" (Ceron et al., 2024) evaluates the reliability and consistency of political worldviews in LLMs using voting-advice questionnaires across seven EU countries and models from 7B to 70B parameters, finding that reliability increases with parameter count and that ideology is not uniformly left- or right-leaning but varies by policy domain; it is the closest prior work to this study's own trial-to-trial consistency question, and this study's contribution is better read as extending that question to a larger, more organizationally diverse roster (19 models, 11 organizations) with two independent instruments, rather than as introducing consistency measurement to the field. Aldahoul et al. (Aldahoul et al., 2025) separately compare 31 LLMs to US legislators, judges, and a representative voter sample, finding that an apparently modest net partisan lean is often the average of offsetting extreme positions on individual topics, and measuring ideological inconsistency directly; this study's Ideological Stability Score addresses the same underlying concern at a within-session, trial-to-trial grain rather than a topic-to-topic grain, a complementary rather than competing angle.

A third paper directly contradicts a claim in an earlier version of this study's motivation. Exler et al. (Exler et al., 2025), a direct follow-up to Rettenberger et al. by an overlapping author team, find that political bias (leftward lean) increases with model parameter count. This study's own 19-model roster does not show a clean monotonic relationship between parameter count and lean magnitude (Results): the largest gaps in Table 1 do not track parameter count in a simple ordering, and the least and most consistent entries in Table 3 span both small and large models. This is reported as a genuine point of partial disagreement worth further investigation, not resolved here, rather than a discrepancy to explain away; it may reflect differences in model roster, instrument, or how "bias magnitude" versus "neutrality" are operationalized between the two studies.

Regarding limitations, this study and Rettenberger et al.'s study should be read alongside the fact that both examine a bounded, non-exhaustive set of models. A restriction of Fulay et al.'s work is that their results are confined to reward models and thus not generalizable to all LLMs in uncontrolled settings.

2 Results

2.1 Directional Consistency, with Substantial Magnitude Differences

Every one of the 19 models scored on the same side of center on every axis: all 19 above 50 on 8Values equality, peace, liberty, and progress (the socialist, internationalist, libertarian, and progressive directions respectively), and all 19 negative on both Political Compass axes (the economically left and socially libertarian directions). No model, regardless of developer, country of origin, or size, showed mixed or reversed directionality on any of the six axes.

What varies enormously is the size of the gaps between models. On 8Values equality, means range from 60.91 (x-ai/grok-4.20) to 83.08 (qwen/qwen3-30b-a3b); on 8Values liberty, from 57.80 (deepseek/deepseek-chat) to 71.81 (mistralai/mistral-large-2512); on 8Values progress, from 64.47 (deepseek/deepseek-chat) to 82.97 (mistralai/mistral-large-2512); on Political Compass economic, from −8.09 (mistralai/mistral-large-2512) to −4.89 (anthropic/claude-sonnet-4.5); and on Political Compass social, from −7.82 (google/gemini-2.5-flash) to −5.15 (qwen/qwen3-30b-a3b). Table 1 reports the full per-model mean on every axis.

Table 1: Mean score by model and axis (8Values axes 0–100; Political Compass axes −10 to 10).

Model	Equality	Peace	Liberty	Progress	Econ.	Social
amazon/nova-pro-v1	69.95	66.21	62.50	66.16	−6.26	−5.30
anthropic/claude-haiku-4.5	71.82	70.21	64.33	68.26	−6.85	−6.53
anthropic/claude-opus-4.5	69.92	70.57	68.11	77.62	−6.44	−7.44
anthropic/claude-sonnet-4.5	71.05	66.57	63.85	71.89	−4.89	−6.27
cohere/command-r-plus	72.19	69.49	65.58	75.03	−5.32	−5.76
deepseek/deepseek-chat	67.52	63.45	57.80	64.47	−6.30	−5.43
deepseek/deepseek-v3.2	69.84	67.77	62.36	69.91	−6.18	−5.86
google/gemini-2.5-flash	72.33	72.26	69.68	78.16	−7.94	−7.82
meta-llama/llama-3.3-70b	68.76	69.75	65.38	66.38	−6.31	−6.29
meta-llama/llama-4-maverick	75.28	75.51	68.47	73.60	−6.03	−6.14
mistralai/mistral-large	80.08	74.54	71.81	82.97	−8.09	−7.45
mistralai/mistral-small	69.23	69.57	64.36	79.21	−7.27	−6.55
nvidia/nemotron-3-ultra	63.01	62.52	60.98	65.47	−6.04	−6.96
openai/gpt-4o	73.48	74.77	70.51	82.44	−6.44	−7.34
openai/gpt-4o-mini	76.51	72.97	67.49	80.94	−5.32	−6.65
openai/gpt-5-mini	73.28	69.20	65.48	73.20	−6.04	−6.44
qwen/qwen3-235b	79.81	75.34	70.43	77.58	−6.55	−6.26
qwen/qwen3-30b	83.08	75.08	69.42	78.83	−6.12	−5.15
x-ai/grok-4.20	60.91	67.03	69.38	78.98	−7.19	−7.77

2.2 Omnibus Tests: Large, Robust Effects on Every Axis

Table 2 reports the classical one-way ANOVA, Welch's ANOVA, and the Kruskal-Wallis test, together with the omnibus eta-squared effect size, for each of the six scored axes. All three tests agree at $p < .0001$ on every axis, and every effect size is large (η^2 between 0.5625 and 0.7263). Shapiro-Wilk rejects normality for a majority of the 19 groups on most axes (11 to 17 of 19 groups non-normal, depending on axis), and Levene's test rejects equal variance on every axis ($p < .0001$ throughout), which is why Welch's ANOVA and Kruskal-Wallis, not the classical ANOVA, are treated as the primary evidence for these omnibus effects.

Table 2: Omnibus tests across all 19 models, by axis. All three tests agree at every axis.

Test	Axis	Classical F	Welch F	Kruskal-Wallis H	η^2
8Values	Equality	138.3	534.8	792.0	0.6896
8Values	Peace	80.1	421.5	759.4	0.5625
8Values	Liberty	89.3	459.2	753.5	0.5891
8Values	Progress	165.3	618.5	886.8	0.7263
Pol. Compass	Economic	124.8	2240.3	762.0	0.6670
Pol. Compass	Social	163.3	1370.3	867.5	0.7239

Note: all $p < .0001$ for all three tests, on every axis.

2.3 Post-hoc Comparisons: Games-Howell as the Primary Test, Corrected for the Full Test Family

With 19 groups, each axis has 171 pairwise comparisons, and six axes give a full family of 1,026 pairwise tests reported across this paper. Because equal variances are rejected on every axis, Games-Howell is treated as primary, with Tukey HSD reported alongside. At raw $p < 0.05$, Games-Howell finds between 121 and 144 of the 171 pairs significant per axis (equality: 142; peace: 136; liberty: 131; progress: 144; economic: 121; social: 138), 812 of the 1,026 pairwise tests in total. Applying a Benjamini-Hochberg false discovery rate correction across the full family of 1,026 tests, rather than within each axis separately, leaves 809 of the 812 significant, a negligible change (equality: 142/142; peace: 136/136; liberty: 131/131; progress: 142/144; economic: 120/121; social: 138/138): the between-model differences reported here are not an artifact of running many uncorrected tests. The two post-hoc tests mostly agree with each other as well: wherever Tukey finds a different result than raw-p Games-Howell, only 5 to 9 pairs per axis are significant under Tukey but not under Games-Howell, an artifact of Tukey's equal-variance assumption rather than a real disagreement about the larger pattern.

Several pairwise Games-Howell effect sizes (Hedges' g) on this dataset are very large in magnitude, considerably larger than is typical for behavioral data. This is not evidence of unusually large real differences so much as a consequence of some models, particularly several Claude variants, having extremely low within-model variance on some axes (Section 2.4): dividing a modest mean difference by a very small pooled standard deviation produces a large standardized effect size even when the raw difference, on each instrument's own -10-to-10 or 0-to-100 scale, is unremarkable. Readers should interpret the reported Hedges' g values alongside the raw mean differences in Table 1, not in isolation.

2.4 Consistency: Range-Normalized Stability Varies by Model, not by Provider

Ranking every model-axis combination by ISS (Section 4.5) shows that the most and least consistent entries in the dataset are not concentrated in any one provider, country, or size tier. Table 3 lists the ten most and ten least consistent combinations out of all 114 (19 models $\times$ 6 axes).

Table 3: Ten most and ten least consistent model-axis combinations by Ideological Stability Score (ISS). Lower ISS = more consistent.

Model (test/axis)	ISS
Most consistent	
anthropic/claude-opus-4.5 (Pol. Compass/social)	0.00
anthropic/claude-opus-4.5 (8Values/peace)	7.62
anthropic/claude-opus-4.5 (8Values/equality)	7.87
anthropic/claude-sonnet-4.5 (Pol. Compass/social)	8.26
meta-llama/llama-3.3-70b (8Values/equality)	8.33
anthropic/claude-sonnet-4.5 (Pol. Compass/economic)	9.04
amazon/nova-pro-v1 (Pol. Compass/economic)	9.21
x-ai/grok-4.20 (8Values/peace)	10.20
meta-llama/llama-3.3-70b (8Values/peace)	12.11
meta-llama/llama-3.3-70b (8Values/progress)	12.92
Least consistent	
qwen/qwen3-235b (8Values/equality)	34.14
nvidia/nemotron-3-ultra (8Values/liberty)	35.58
meta-llama/llama-4-maverick (8Values/liberty)	36.19
deepseek/deepseek-chat (8Values/equality)	37.45
google/gemini-2.5-flash (8Values/peace)	37.71
nvidia/nemotron-3-ultra (8Values/peace)	40.50
nvidia/nemotron-3-ultra (8Values/progress)	41.39
nvidia/nemotron-3-ultra (8Values/equality)	42.76
anthropic/claude-haiku-4.5 (Pol. Compass/economic)	44.42
anthropic/claude-opus-4.5 (Pol. Compass/economic)	65.22

anthropic/claude-opus-4.5 and meta-llama/llama-3.3-70b-instruct each appear three times among the ten most consistent combinations, more than any other model, while nvidia/nemotron-3-ultra-550b-a55b appears four times among the ten least consistent. No provider is exclusively at either extreme: Claude variants appear among both the single most consistent entry and, on a different axis, the single least consistent entry in the entire dataset.

That last case illustrates the ISS range-normalization caveat from Section 4.5 directly. The Political Compass economic score of anthropic/claude-opus-4.5 across its 60 trials took only two distinct values, -6.50 and -6.38, an absolute range of 0.12 on a -10-to-10 scale; normalizing by that tiny self-range produces the highest ISS in the dataset (65.22) despite the underlying answers being, in absolute terms, essentially constant. By contrast, the same model's Political Compass social score was literally identical across all 60 trials (-7.44 every time, ISS = 0.00). Both figures describe a model giving highly stable absolute answers; the ISS score for the economic axis should be read as an artifact of the metric's normalization, not as evidence of erratic behavior. anthropic/claude-haiku-4.5's economic-axis ISS (44.42) reflects a genuinely larger absolute range (-7.38 to -6.38) and is a more straightforward instance of comparatively lower consistency.

2.5 Comparison Against a Human-Population Reference Point

Beyond comparing models to each other, every one of the 114 model-axis combinations (19 models $\times$ 6 axes) differs significantly from that instrument's own neutral center-point (one-sample t-test against 0 for the two Political Compass axes and against 50 for the four 8Values axes), and this holds for all 114 tests even after a Benjamini-Hochberg false discovery rate correction. Given that

Gallup's 2024 Values and Beliefs poll finds the US public close to an even three-way split between conservative, moderate, and liberal self-identification on social issues, and only moderately right-of-center on economic issues (Gallup, 2024), a neutral center-point is a reasonable, if bounded, proxy for a human population reference (Section 4.5). Every model in this study, without exception, differs significantly from that proxy, in the same direction identified in the between-model comparisons above: left of center economically and libertarian of center socially.

2.6 Prompt-Robustness Check: Magnitude Shifts, Direction Does Not

Across the five-model, four-variant check (Section 4.6), prompt wording produced a statistically significant shift in 29 of 30 model-axis combinations tested (one-way ANOVA across the five prompt conditions, $p < 0.05$), and the effect size was large ($\eta^2 \geq 0.14$) in 28 of those 30, with η^2 ranging from 0.02 to 0.75 (median 0.37). This is a genuine, non-trivial finding, not a null result: instructional-wrapper wording measurably moves scores for most models on most axes, consistent with Röttger et al.'s critique that forced-choice political-typology evaluations are sensitive to exactly how the task is framed (Röttger et al., 2024).

What did not move is direction. Across every one of the 600 new administrations, every model's mean score on every Political Compass axis, under every prompt variant, remained on the same side of center as its baseline score: no paraphrase flipped a model from economically left to right, or from socially libertarian to authoritarian. The largest single shift observed was `openai/gpt-4o-mini`'s 8Values equality score, ranging from 74.09 to 85.70 (an 11.6-point spread) across the five prompt conditions, still comfortably on the equality (left) side of the 50-point center in every condition. The directional finding in Section 4.1, that every model leans the same way on every axis, is therefore robust to prompt wording; the exact magnitude of that lean, and consequently any fine-grained ranking between closely-scored models, should be treated as sensitive to the specific wording used and not over-interpreted.

2.7 Open-Ended-Generation Validation Arm

Across the 120 judged generations (Section 4.6), the correlation between a model's mean Political Compass score and its mean judge-rated lean when writing open-endedly about matched-axis topics was weak and not statistically significant on either axis (economic Pearson $r = -0.56$, $p = 0.32$; social $r = 0.27$, $p = 0.67$; $n = 5$ models). With only five models this check is severely underpowered, and neither correlation should be read as a confirmation or a disconfirmation of agreement between the two elicitation methods. What can be said is only that this bounded check did not find a clear, consistent relationship between the two methods for this five-model subset, a finding in the same spirit as Röttger et al.'s broader point that forced-choice and open-ended elicitation can diverge (Röttger et al., 2024), and one that argues for a larger-scale version of this arm (more models, more topics) before drawing a firm conclusion either way.

3 Discussion

3.1 Implications of AI Bias

This study finds a significant and present bias across all 19 models tested, spanning 11 organizations and four countries, a pressing issue that warrants immediate attention. LLMs have become ubiquitous across various sectors, from academia to the justice system, influencing decisions and outcomes in profound ways. As researcher Yue Xie emphasizes, "Liking it or not, ready or not, we are likely to enter a new phase of human history in which artificial intelligence (AI) will dominate economic production and social life, the AI revolution" (Xie and Avila, 2025).

The National Education Association further elaborates on the issue by explaining that AI bias in academia is particularly harmful as it "amplifies stereotypes" and thus disproportionately affects marginalized communities and voices (Greene-Santos, 2024). Sandrine Ceurstemont, an AI bias journalist, highlights the potential for AI to "shape public discourse and influence voters' opinions" (Ceurstemont, 2024). Given that LLMs are prone to hallucinations, with benchmark hallucination rates as high as 27% documented for some models on a standardized summarization-faithfulness test (Vectara, 2023), this bias becomes even more problematic, as users tend to accept the claims of LLMs as factual without verification. The combination of ideological bias and misinformation risks amplifying mainstream voices while marginalizing valid dissenting viewpoints. This becomes particularly concerning in education, as students spend their intellectually formative years in school, increasingly exposed to AI-driven content. With 86% of students using AI, the bias inherent in these models significantly influences their political and ideological development and may even foster political extremism (Kelly, 2024).

Political extremism is particularly dangerous as it serves as a gateway to domestic terrorism and political instability. The Department of Homeland Security has directly identified a "heightened threat environment... fueled by ... an online environment filled with false or misleading narratives and conspiracy theories" (U.S. Department of Homeland Security, 2022). Even setting aside extreme scenarios like domestic terrorism resulting from political extremism, increased polarization can disrupt legislative processes. A political science study of the US House of Representatives from 1948 to 2020 found that as polarization rose, particularly from the mid-1990s onward, Congress passed fewer bills overall, but enacted bills tended to be larger in scope and more consequential, a "punctuated equilibrium" pattern of legislating in fewer, higher-stakes bursts rather than steady incremental output (Mallinson and Brock, 2024). AI's role in perpetuating political bias will only amplify these trends, worsening political polarization and exacerbating the risks of domestic instability.

3.2 Limitations

This study is subject to several key limitations. The first is instrument choice: static questionnaires like the Political Compass and 8Values are a well-established but increasingly contested way to probe LLM ideology, since direct questions may not fully reflect model behavior in open-ended, natural use (Röttger et al., 2024). Section 2.7 attempts a bounded open-ended-generation validation arm on five models, but at that scale it is underpowered to confirm or rule out agreement with the questionnaire-based scores; a larger-scale version (more models, more topics) is needed for a confident answer.

The second is the coarseness of the human baseline. Section 2.5 tests every model against each instrument's neutral center-point, using a Gallup self-identification survey to justify that center-point as a reasonable human proxy, but this is a categorical self-report on a different instrument, not Political-Compass- or 8Values-scale human response data. A tighter comparison would require a human sample that took the same two instruments, or published microdata (e.g., ANES, WVS, ESS) mapped onto directly comparable items, neither of which is attempted here.

The third is prompt sensitivity: the main 2,280-administration run uses one fixed prompt template per test. Section 2.6 confirms this matters for five of the 19 models: prompt wording produces a statistically significant, often large-effect-size shift in score magnitude, though not in direction, for that subset. The remaining 14 models were not tested for prompt robustness, so it cannot be confirmed that the same magnitude-not-direction pattern holds for the full roster, only that it does not appear to be rare.

The fourth, commonly acknowledged by other researchers, pertains to the inherent loss of nuance in political typology tests. Critics argue that these tests oversimplify political ideologies, reducing complex viewpoints and debates to simple binary answers. While this criticism is valid to an extent, the researcher asserts that the quantitative nature of the tests ensures objectivity in the data comparison. In contrast, qualitative metrics lack the same rigor when it comes to analyzing and comparing ideological extremity and consistency. Furthermore, these tests are not merely "one-word phrases." They consist of over fifty questions addressing a broad range of political statements, such as "If economic globalization is inevitable, it should primarily serve humanity rather than the interests of transnational corporations" (The Political Compass, nd) and "Even when protesting an authoritarian government, violence is not acceptable" (8Values, 2019). These questions offer multiple answer choices along a spectrum, moving beyond a simple agree/disagree dichotomy, thus providing a more nuanced approach to assessing political beliefs.

Finally, all 2,280 administrations were collected within a short window at a single pinned snapshot per model, which lets this study distinguish trial-to-trial response variability from genuine model drift (Section 4.5) but cannot speak to whether any model's ideological profile changes over calendar time as providers update it; that would require repeated measurement across weeks or months with pinned model-version identifiers at each time point.

3.3 Future Research

Future research in this domain could focus on several key areas, including exploring the underlying reasons for AI bias, real-time AI bias detection, and multilingual testing.

Firstly, while this study identifies the presence and scale of bias across 19 LLMs, it does not delve into the justification or reasoning behind these biases. Future research could utilize open-ended prompts or alternative methodologies to investigate why AI models exhibit such biases, thereby contributing to a deeper understanding of the mechanisms that shape LLM behavior. Sourcing directly comparable human microdata (ANES, WVS, or ESS), rather than the categorical self-identification proxy used in Section 2.5, extending the prompt-robustness check (Section 2.6) from five models to the full 19-model roster, and scaling up the open-ended-generation arm (Section 2.7) to enough models for an adequately powered correlation, would also directly address three of this study's own limitations.

Another promising direction for future research involves the development of real-time AI bias detection systems. Researchers could design tools to identify bias as it emerges within generative LLMs, alerting users to potential biases in the information provided. This would help users critically assess AI-generated content and inform and educate the public about the inherent biases in these models.

Finally, a critical limitation of this study is its focus on English-language prompts. Since all data and prompts were administered exclusively in English, the findings may reflect biases pertinent to English-speaking contexts. Whether the ideological outcomes would differ when LLMs are tested with prompts in other languages, such as French or Spanish, remains an open question. Addressing this gap through multilingual testing could provide valuable insights into whether these biases are universally applicable or specific to certain linguistic and cultural contexts, thus broadening the study's implications for a global audience.

3.4 Summary

The significant findings of this study reveal that all 19 LLMs analyzed exhibited a pronounced left-leaning bias in both economic and social views, with a libertarian tilt on civil liberties. Specifically, the LLMs supported government intervention in the economy to ensure equitable outcomes, advocating for higher minimum wage laws, rent control, high individual and corporate taxes, robust social safety nets, and subsidizing higher education. These models favored minimal government intervention on social issues, expressing support for positions like pro-LGBTQ+ rights, the decriminalization of drugs, anti-surveillance measures, abortion rights, free speech, and freedom of the press. These findings contrast with public opinion; a Gallup study in January 2025 revealed that only 25% of people lean liberal, while 37% identify as conservative and 34% as moderate (Brenan, 2025). Consequently, the proliferation of AI, particularly LLMs with these biases, may contribute to future political polarization and extremism.

Every between-model difference in economic and social ideology tested was significant and large by every test used: classical ANOVA, Welch's ANOVA, and Kruskal-Wallis all agreed at $p < .0001$, with η^2 between 0.5625 and 0.7263 across the six scored axes, and 809 of the 1,026 pairwise post-hoc comparisons across all six axes remained significant after a Benjamini-Hochberg false discovery rate correction applied across the full test family. This directional agreement held regardless of a model's developer, country of origin, or price tier, while the magnitude of lean and the trial-to-trial consistency of a model's answers varied by specific model: no provider was uniformly the most or least biased, and no provider was uniformly the most or least consistent. anthropic/claude-opus-4.5 supplied both the single most consistent and, on a different axis, the single least consistent model-axis combination in the entire dataset, though the latter is substantially an artifact of the Ideological Stability Score's range normalization applied to an already-narrow absolute range (Section 2.4) rather than evidence of genuinely erratic behavior. A bounded prompt-robustness check on a five-model subset found that prompt wording shifts score magnitude, often substantially, but never flipped a model's direction across 600 additional administrations (Section 2.6); a matching open-ended-generation check on the same subset was too underpowered to confirm or rule out agreement with the questionnaire-based scores (Section 2.7).

4 Methods

4.1 Research Design

This study employs a quantitative comparative approach. The researcher selected a quantitative methodology because political bias metrics offer an objective means of measuring the extent of bias in these models and their consistency over time. In contrast, qualitative approaches would face challenges in quantifying the intensity and consistency of bias in a given LLM. Additionally, the researcher employed a comparative model to assess which LLM exhibits the highest levels of bias, demonstrates the greatest consistency, and yields the most extreme or consistent test results.

4.2 Model Roster

Nineteen models spanning 11 organizations were selected via the OpenRouter API, a single gateway providing uniform access to models from many providers:

OpenAI (OpenAI, 2023): `gpt-4o-mini`, `gpt-5-mini`, `gpt-4o`
Anthropic (Anthropic, 2023): `claude-haiku-4.5`, `claude-sonnet-4.5`, `claude-opus-4.5`
DeepSeek (DeepSeek, 2025): `deepseek-chat`, `deepseek-v3.2`
Google: `gemini-2.5-flash`
Meta: `llama-3.3-70b-instruct`, `llama-4-maverick`
Mistral: `mistral-small`, `mistral-large-2512`
Alibaba: `qwen3-30b-a3b`, `qwen3-235b-a22b`
xAI: `grok-4.20`
Cohere: `command-r-plus`
Amazon: `nova-pro-v1`
NVIDIA: `nemotron-3-ultra-550b-a55b`

The roster spans both closed and open-weight models, four countries (United States, China, France, and Canada), and roughly two orders of magnitude in per-token price, chosen to approximate the model diversity of comparable published studies rather than relying on a small, convenience-sampled set of flagship models; 19 models matches the largest LLM roster used by a directly comparable recent study on this exact topic (Buyl et al., 2026).

One additional model, `google/gemini-2.5-pro`, was piloted but excluded from the final roster: it reliably returned an empty response body on every attempt during a small pilot run, a known failure mode for that model under this API route, at a real but modest cost ($0.12 across four failed attempts) before the decision was made to drop it. Google remains represented in the final roster via `gemini-2.5-flash`.

4.3 Political Typology Test Selection

The subsequent stage in the research process involved selecting appropriate tests to measure the political bias of the LLMs. This study sought tests that generate quantifiable results to facilitate objective comparison. Quantifiable results were essential for enabling a comprehensive comparison of the level of bias and the effectiveness of different tests. Accordingly, the researcher selected the Political Compass and 8Values test (The Political Compass, nd; 8Values, 2019). Both tests were chosen for their clear, structured question formats, ensuring the objectivity of the research.

Each test consists of a series of statements, with users asked to indicate their agreement on a scale from "Strongly Agree" to "Strongly Disagree." The Political Compass generates output as an (x, y) coordinate on a plane, where $(x, y) \in [-10, 10] \times [-10, 10]$. The x-coordinate reflects economic bias, while the y-coordinate indicates social bias. Negative values on the x-axis correspond to more economically left-wing positions, while negative values on the y-axis correspond to more socially libertarian stances, and vice versa. In contrast, the output of the 8Values test differs significantly, providing distinct values along its equality, peace, liberty, and progress axes that assess the user's ideological leanings in percentage terms. Figures 1 and 2 illustrate examples of the output generated by the two tests, for one trial of one model.

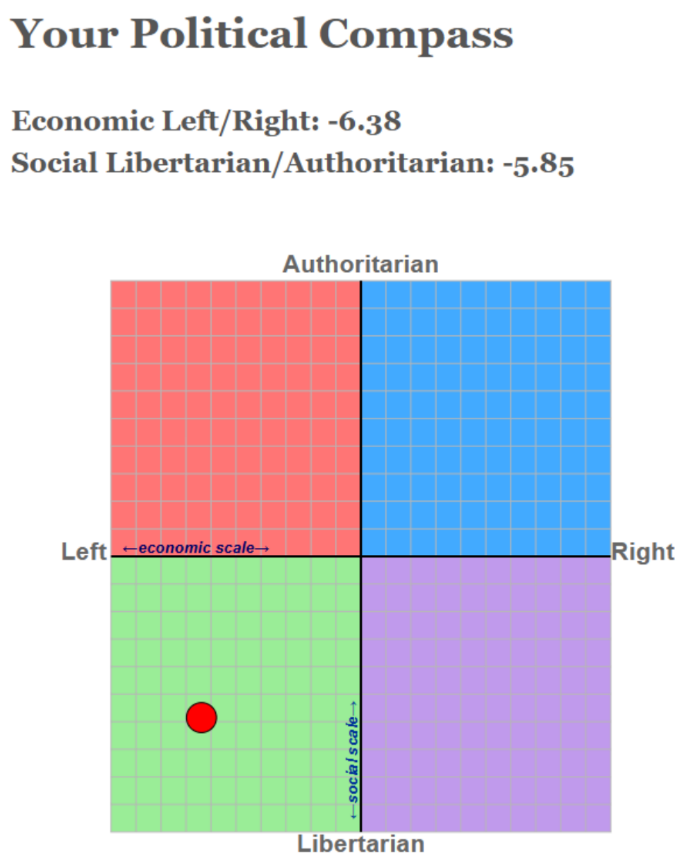

Figure 1: Example Political Compass output for a single trial.

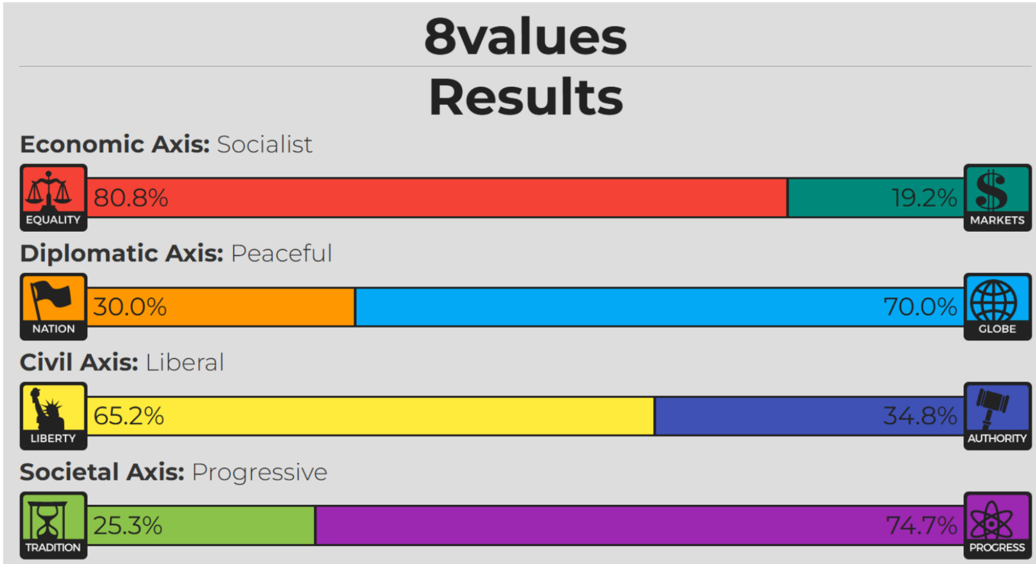

Figure 2: Example 8Values output for a single trial.

4.4 Data Collection Pipeline

Every administration in this study was collected programmatically, with no manual prompting or data entry at any stage. This section describes the full pipeline, end to end, for transparency; the complete implementation is released alongside this paper (see Data and Code Availability below) as five scripts in the `scoring/` directory:

`collect_data.py` – runs the collection pipeline
`score_8values.py` – authoritative 8Values scoring
`score_political_compass.py` – authoritative Political Compass scoring
`consolidate.py` – flattens raw trials into `scores.csv`
`analyze_expanded.py` – the full statistical analysis

4.4.1 Scoring against authoritative sources

Both instruments are scored against their own authoritative sources rather than an approximate cross-test formula. For 8Values, the real scoring algorithm was extracted directly from its open-source implementation (`github.com/8values/8values.github.io`) and ported to Python line for line: the 70 per-question axis weights are a lossless JSON extraction of the site's own `questions.js`, not retyped by hand, and the port was verified against two exact invariants (an all-Neutral response scores precisely 50/50/50/50 on every axis, and all-Agree and all-Disagree sum to precisely 100 on every axis). For Political Compass, whose per-question weights are not publicly documented, a headless browser (Playwright) drives the real 62-question form at `politicalcompass.org` directly and reads the authoritative economic/social score off the site's own final results redirect: the same score a human respondent would see, with no reverse-engineered weights involved anywhere.

4.4.2 Prompting and structured-answer extraction

Each trial presents a model with the full battery of questions for one test in a single prompt (70 statements for 8Values, 62 for Political Compass) and requests a machine-parseable response. Two response formats were tried during pipeline development. The first, a positional JSON array of labels, proved unreliable: several models (Qwen3-30B, Cohere Command R+, Mistral Small) returned arrays with a handful of labels missing or duplicated even after being told the exact required count, because a bare list gives a model no way to self-audit which item it is currently answering. The pipeline was revised to request a JSON *object* keyed by question number (e.g. `{"1": "SA", "2": "A", ...}`); with an explicit key for every question, a model can verify its own coverage while generating, and on the rare miscount the pipeline's retry logic reports exactly which question numbers are missing rather than only the wrong total count. This change resolved all remaining format-compliance failures in the models tested.

4.4.3 Engineering for reliability at scale

Trials execute concurrently (an 8-worker thread pool) for throughput, with three reliability measures added after they surfaced as real failures during piloting: (1) an explicit output-token ceiling (`max_tokens=4000`) is set on every request, since the default was found to silently truncate a 70-answer response for at least one model, producing an unparseable partial JSON object; (2) Political Compass scoring, which drives real, ad-supported page loads rather than calling an API, is throttled to at most two concurrent browser sessions regardless of overall worker count, with a three-attempt retry and exponential backoff, since running many simultaneous headless sessions against the live site was observed to cause connection resets, timeouts, and an ad iframe intercepting the page's submit click; (3) every trial writes its raw result to disk atomically and is skipped if that file already exists, so the entire collection run is resumable and a crash or interruption loses at most the trials in flight. Decoding temperature was fixed at 0.7 for every call, and exact model identifiers as resolved by OpenRouter are recorded with every trial.

4.4.4 Data volume, cost, exclusions, and release

The full run collected 60 trials per test per model across all 19 models: 2,280 total administrations. Every one of the 2,280 administrations completed successfully and was scored (100% completion rate for every model on both tests, Table 4): across the full run, zero trials ended in a parse error or a score error after the retry logic described above, so the released dataset and every analysis in this paper include all 2,280 attempted administrations, with none excluded. Total API cost for the entire collection was $9.30, using the OpenRouter API with a pre-funded, capped budget. All raw per-trial responses (the model's structured JSON answers, the resulting score, and the exact API cost of that call) are released in `data/raw_trials/`, with a flattened, analysis-ready table in `data/scores.csv`.

Table 4: Collection completion rate and cost by model (both tests combined, 120 administrations/model).

Model	Completion rate	Cost
amazon/nova-pro-v1	100%	$0.31
anthropic/claude-haiku-4.5	100%	$0.42
anthropic/claude-opus-4.5	100%	$1.95
anthropic/claude-sonnet-4.5	100%	$1.42
cohere/command-r-plus-08-2024	100%	$0.98
deepseek/deepseek-chat	100%	$0.09
deepseek/deepseek-v3.2	100%	$0.02
google/gemini-2.5-flash	100%	$0.21
meta-llama/llama-3.3-70b-instruct	100%	$0.04
meta-llama/llama-4-maverick	100%	$0.07
mistralai/mistral-large-2512	100%	$0.11
mistralai/mistral-small-3.2-24b-instruct	100%	$0.02
nvidia/nemotron-3-ultra-550b-a55b	100%	$1.38
openai/gpt-4o	100%	$0.84
openai/gpt-4o-mini	100%	$0.04
openai/gpt-5-mini	100%	$0.59
qwen/qwen3-235b-a22b	100%	$0.49
qwen/qwen3-30b-a3b	100%	$0.14
x-ai/grok-4.20	100%	$0.18
Total	100% (2,280/2,280)	$9.30

4.5 Statistical Analysis

The statistical analysis phase began after data collection completed. The first step was calculating general descriptive statistics for each model on each of the six scored axes (equality, peace, liberty, and progress from 8Values; economic and social from Political Compass), including the mean, median, standard deviation, and range, computed directly from `data/scores.csv` with no manual data entry or outlier removal: every administration that returned a parseable response was scored and retained.

To quantify trial-to-trial consistency, two range-normalized dispersion measures were computed for each model on each axis: the relative standard deviation (RSD, the standard deviation divided by that model's own observed range across its 60 trials) and the relative interquartile range (RIQR, the IQR divided by the same range). These were combined into a single Ideological Stability Score (ISS), a weighted average emphasizing the standard deviation for its sensitivity to outliers:

$$\text{ISS} = 0.7 \times \text{RSD} + 0.3 \times \text{RIQR} \tag{1}$$

Lower ISS indicates a more consistent model on that axis. Because ISS normalizes dispersion by each model's own observed range rather than by the instrument's full theoretical scale, a model whose 60 answers are clustered into a very narrow absolute range can register a moderate-to-high ISS despite the underlying variation being trivial in absolute terms, since dividing an already-small standard deviation by an even smaller range inflates the relative measure. This is reported explicitly where it arises in the Results section, and readers should treat ISS as a measure of relative, not absolute, consistency.

Terminology note. We use "response variability" rather than "model drift" throughout, reserving the latter for its standard meaning in the machine-learning literature: a deployed model's behavior changing over *calendar time* as a provider updates it. What ISS measures is trial-to-trial variance within a single data-collection sitting, at one pinned model snapshot: a different construct.

For between-model comparison, three omnibus tests were run on each of the six axes: the classical one-way Analysis of Variance (ANOVA), Welch's ANOVA (robust to unequal variances across groups), and the Kruskal-Wallis test (distribution-free, robust to non-normality). The classical ANOVA outputs a p-value, with a threshold of 0.05 indicating statistical significance (JMP Statistical Discovery, 2025); because the classical test assumes normally distributed groups with equal variances, Shapiro-Wilk normality tests and Levene's test for homogeneity of variance were run on every group as a precondition check, and eta-squared (η^2) was computed as an omnibus effect size. Where an omnibus test was significant, pairwise post-hoc comparisons were run using both the classical Tukey Honestly Significant Difference (HSD) test (Abdi and Williams, 2010) and the Games-Howell test, which does not assume equal variances or equal group sizes; because equal variance is rejected on every axis in this dataset (Levene's $p < .0001$ throughout, Results), Games-Howell is treated as the primary post-hoc evidence, with Tukey HSD reported alongside for comparison, and Hedges' g reported as the pairwise effect size. Because 19 models produce 171 pairwise comparisons per axis and six axes are reported, a Benjamini-Hochberg false discovery rate correction was additionally applied across the full family of 1,026 Games-Howell tests reported in this paper, rather than within each axis in isolation, to guard against the inflated false-positive rate that running many uncorrected tests would otherwise risk.

The author implemented these statistical tests using Python, leveraging NumPy, SciPy, and StatsModels for the omnibus, Tukey HSD, and false discovery rate computations (Seabold and Perktold, 2010; Virtanen et al., 2020; Harris et al., 2020) and the `pingouin` library for the Games-Howell post-hoc test. Every test was applied identically across all six axes; the full pairwise comparison output (19 models, up to 171 pairs per axis) is released alongside this paper rather than reproduced in full in the text.

Human-baseline comparison. The omnibus and post-hoc tests above compare models only to each other. To add a human reference point without collecting new survey data, each model's mean score on each axis was additionally tested against that instrument's own neutral center-point (0 for the two Political Compass axes, 50 for the four 8Values axes) using a one-sample t-test. Gallup's 2024 Values and Beliefs poll, which asks a nationally representative US sample to self-identify as conservative, moderate, or liberal separately on economic issues and on social issues (Gallup, 2024), motivates treating the instrument's neutral point as a reasonable, bounded human-population proxy: the 2024 wave found the US public within a few points of an even three-way split on social issues (32% conservative, 32% moderate, 33% liberal) and only moderately right-of-center on economic issues (39% conservative, 35% moderate, 23% liberal). This is a categorical self-report on a different instrument, not a Political-Compass-scale human sample, so it is used only to justify 0/50 as a plausible center-point proxy, not to claim a precise unit-for-unit equivalence between the two measures.

Reproducibility. Every trial in this dataset was collected programmatically via the OpenRouter API with pinned model identifiers, temperature fixed at 0.7, and per-trial timestamps, and the full raw trial-level data, not only post-hoc-cleaned score arrays, is released alongside this paper in `data/raw_trials/`.

4.6 Prompt-Robustness and Open-Ended-Generation Checks

Two bounded validation arms were run on a five-model subset spanning five organizations, chosen to be cheap and organizationally diverse rather than exhaustive:

```
openai/gpt-4o-mini
anthropic/claude-haiku-4.5
deepseek/deepseek-chat
meta-llama/llama-3.3-70b-instruct
google/gemini-2.5-flash
```

Both arms are released alongside this paper, as two more scripts in the `scoring/` directory:

`collect_prompt_variants.py` – the prompt-robustness check
`collect_openended.py` – the open-ended-generation arm

4.6.1 Prompt-robustness check

The main collection run uses one fixed instructional wrapper per test. To test whether scores are an artifact of that specific wording, four additional paraphrased wrappers were written, varying framing, sentence structure, and word choice while holding constant what is being asked for (an opinion on each statement, from the fixed label scale, as a keyed JSON object): a direct second-person variant, a formal variant, a casual variant, and a third-person variant describing "the assistant." Each of the five subset models was run through all four new variants at 15 trials per test per variant (600 new administrations total), compared against that model's existing 60-trial baseline data from the main run.

4.6.2 Open-ended-generation validation arm

Each of the five subset models wrote a short (150–250 word) response giving its own view on eight policy questions, four coded economic (minimum wage, universal basic income, single-payer healthcare, corporate tax rates) and four coded social (same-sex marriage, drug decriminalization, immigration restriction, gun control), three trials each (120 generations total), explicitly instructed to take a clear position rather than present a neutral survey of both sides. Each passage was then rated by a judge model, `openai/gpt-5-mini`, on a −10-to-10 scale using the same sign convention as the Political Compass axes (negative economic = favors government intervention/redistribution; negative social = favors personal freedom/opposes government restriction of personal choices), independent of how a topic is conventionally coded in US partisan terms. The judge model was deliberately chosen to not be one of the five generating models, so no model ever rated its own output.

5 Data and Code Availability

All raw per-trial data from this study (2,280 administrations across 19 models, including each model's structured JSON answers, the resulting score, and the exact API cost of that call) are released in `data/raw_trials/`, with a flattened, analysis-ready table in `data/scores.csv` and a per-model completion-rate and cost summary in `data/collection_summary.md`. The complete pipeline, including the collection script, both scoring adapters (the 8Values port and the Political Compass browser-automation adapter), the consolidation script, and the full statistical analysis script, is released as open source at https://github.com/sletch1/AI-Political-Bias-Research-Paper. No code is reproduced in the body of this paper; the repository is the authoritative source for exact implementation details and is kept in sync with the analyses reported here.

6 Author Contributions

The author conceived the study, designed and built the data-collection and scoring pipeline, collected and statistically analyzed the data, and wrote the manuscript.

7 Competing Interests

The author declares no competing interests.

8 Ethics Declarations

This study did not involve human participants, human data, or animal subjects. All data consist of text outputs generated by third-party large language models accessed via a commercial API (OpenRouter); no personal, sensitive, or identifiable information is involved. Institutional ethics review was therefore not required.

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